

Development of a Wireless Active Personal Dosimeter for Interventional Radiology

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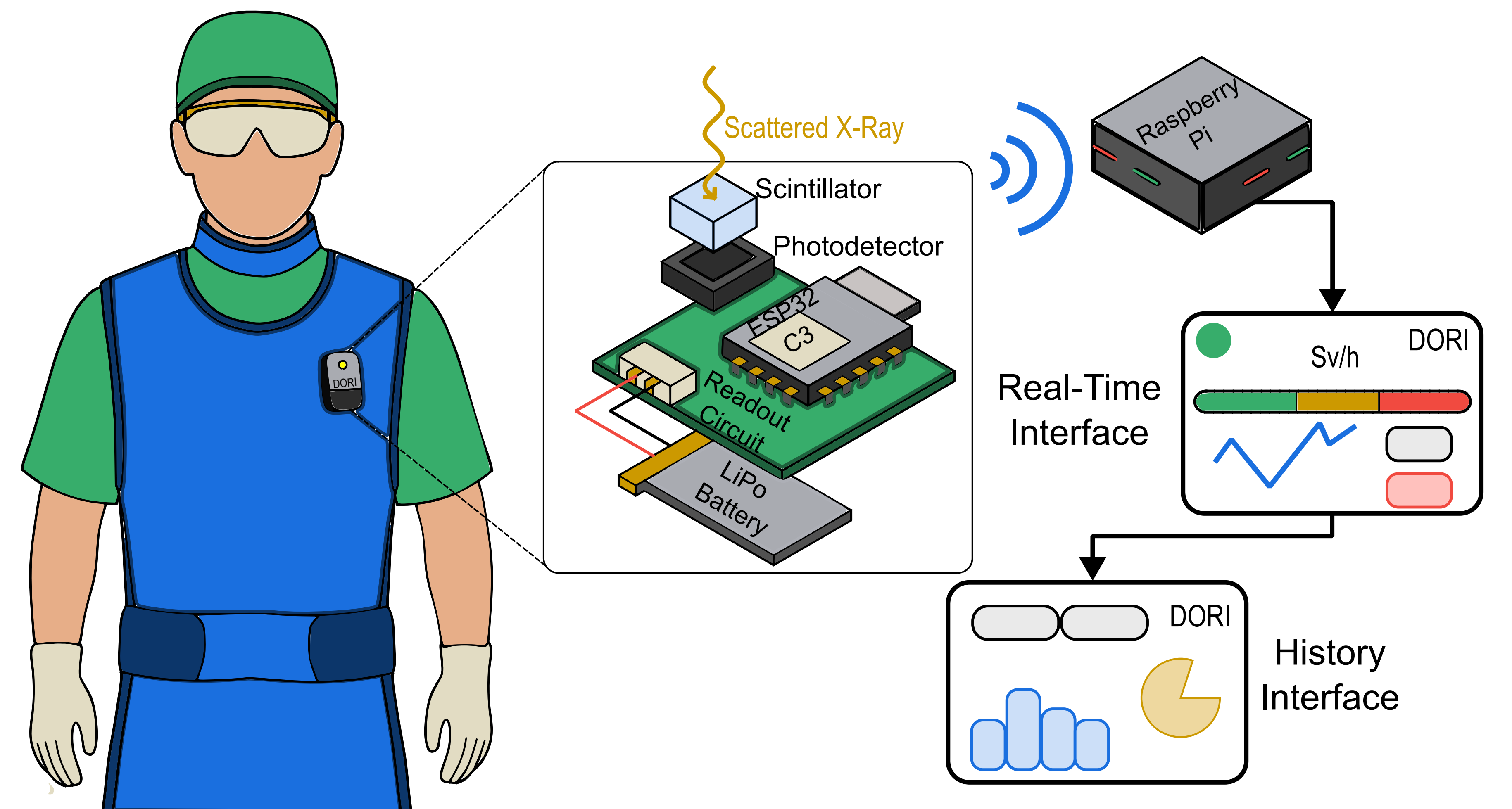
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ABSTRACT

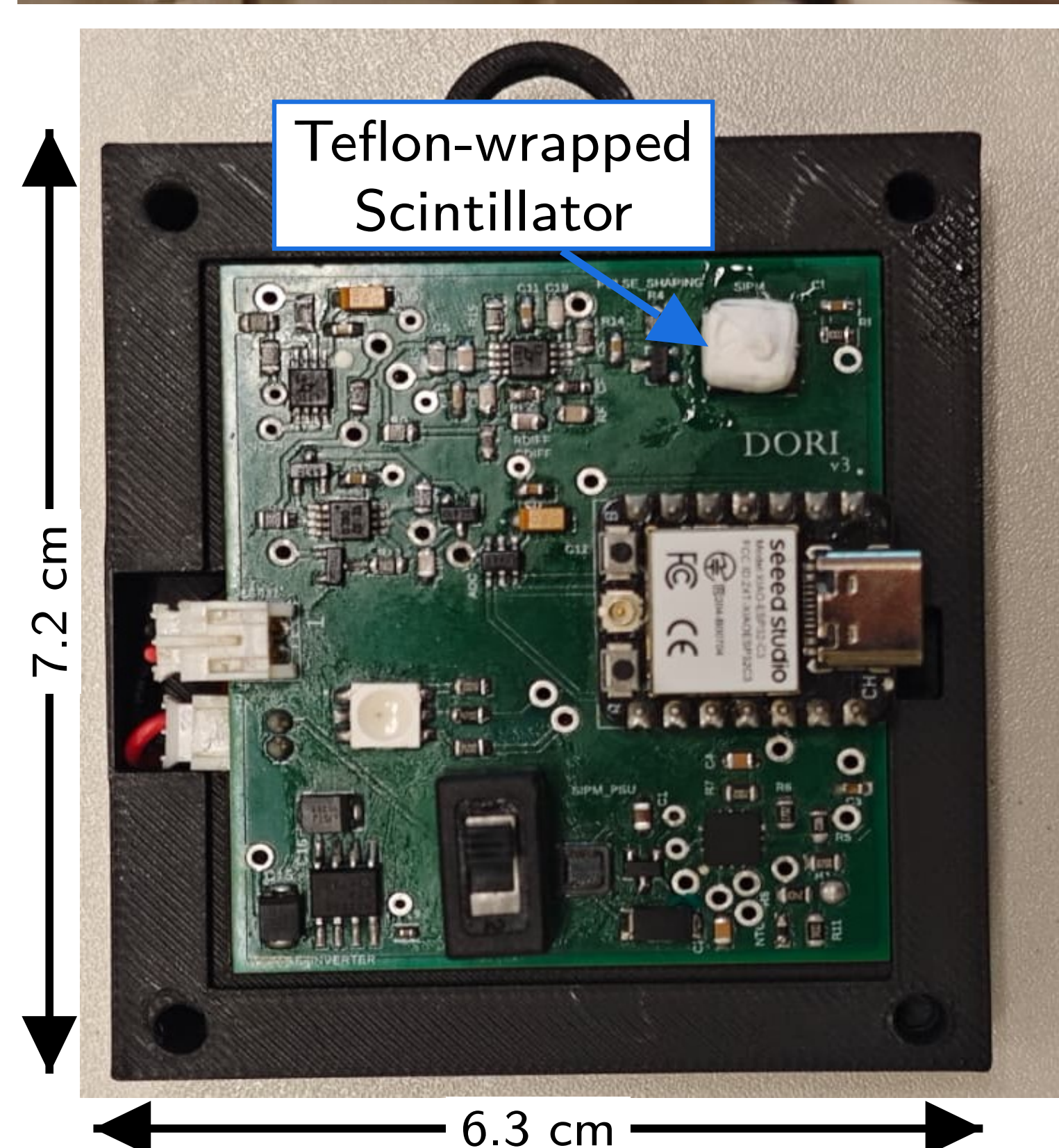
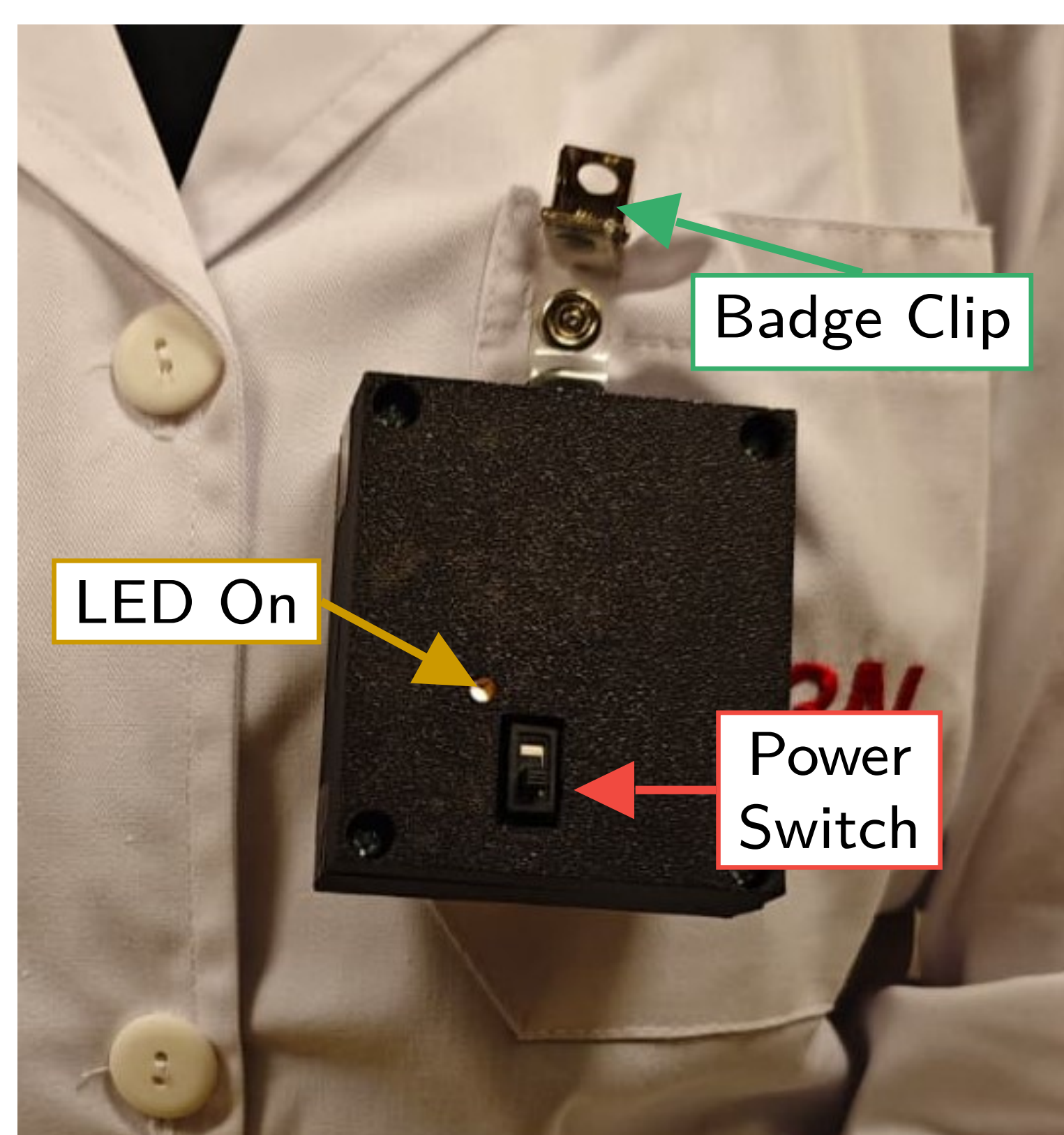
Interventional radiology is the medical specialty subject to the highest occupational exposure to ionising radiation. Monitoring still relies on passive dosimeters, which report dose retrospectively, so exposure accidents cannot be prevented.

This work presents DORI (*DO*simetro para *R*adiologia de *I*ntervenção), a wireless active personal dosimeter for real-time monitoring of scattered radiation. The sensing element is a $5 \times 5 \times 5 \text{ mm}^3$ BC-408 plastic scintillator coupled to a Broadcom SiPM with a $6 \times 6 \text{ mm}^2$ active area. The readout electronics is controlled by an ESP32-C3 powered by a rechargeable Li-Po battery. Energy calibration was performed with bremsstrahlung endpoints and the ^{241}Am photopeak, with the ^{57}Co Compton edge as an independent validation point. The dose response followed ISO 4037 narrow spectra in terms of $H_p(10)$. The prototype is compact, fully wireless and unobtrusive to wear.

METHODS

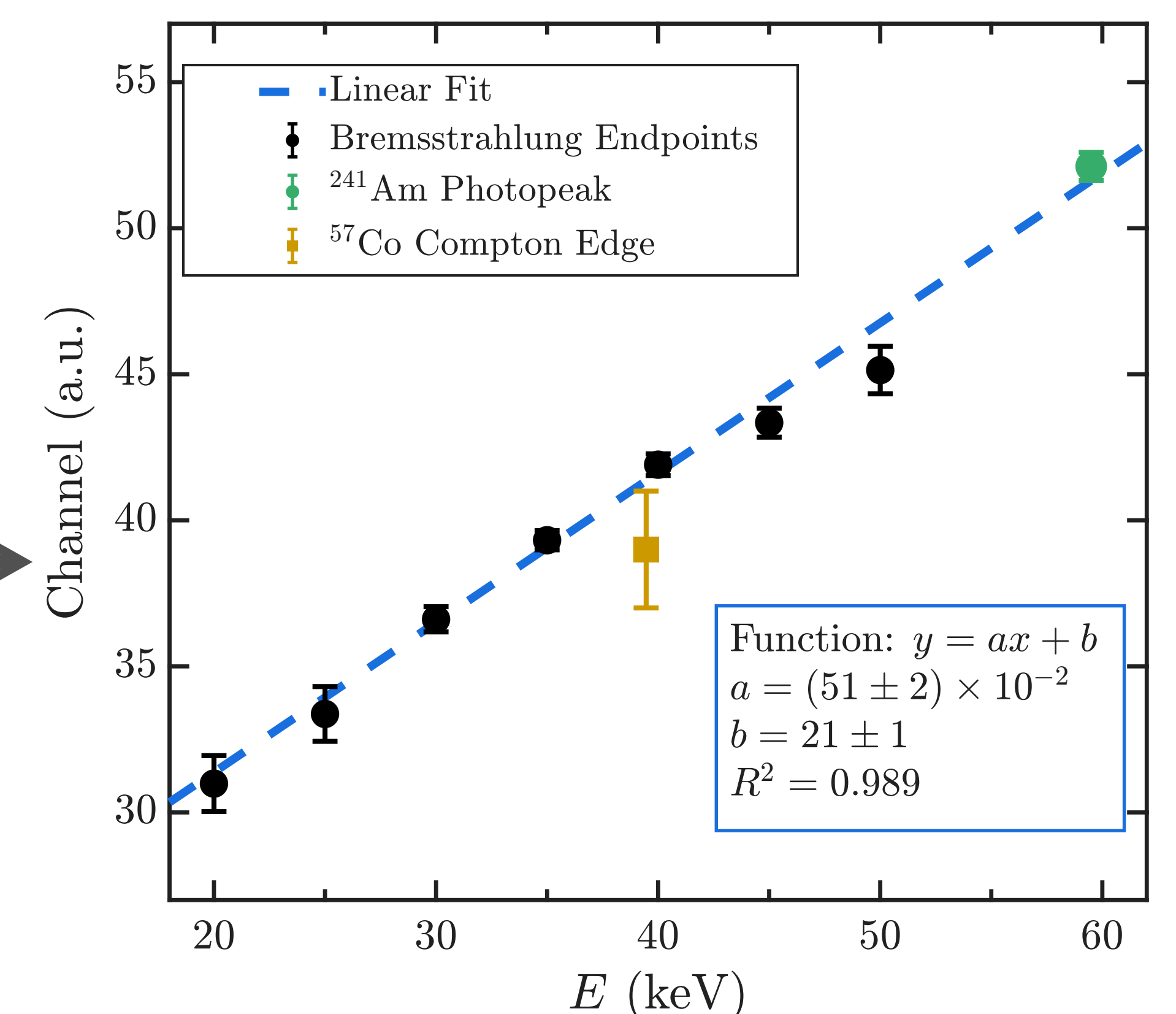
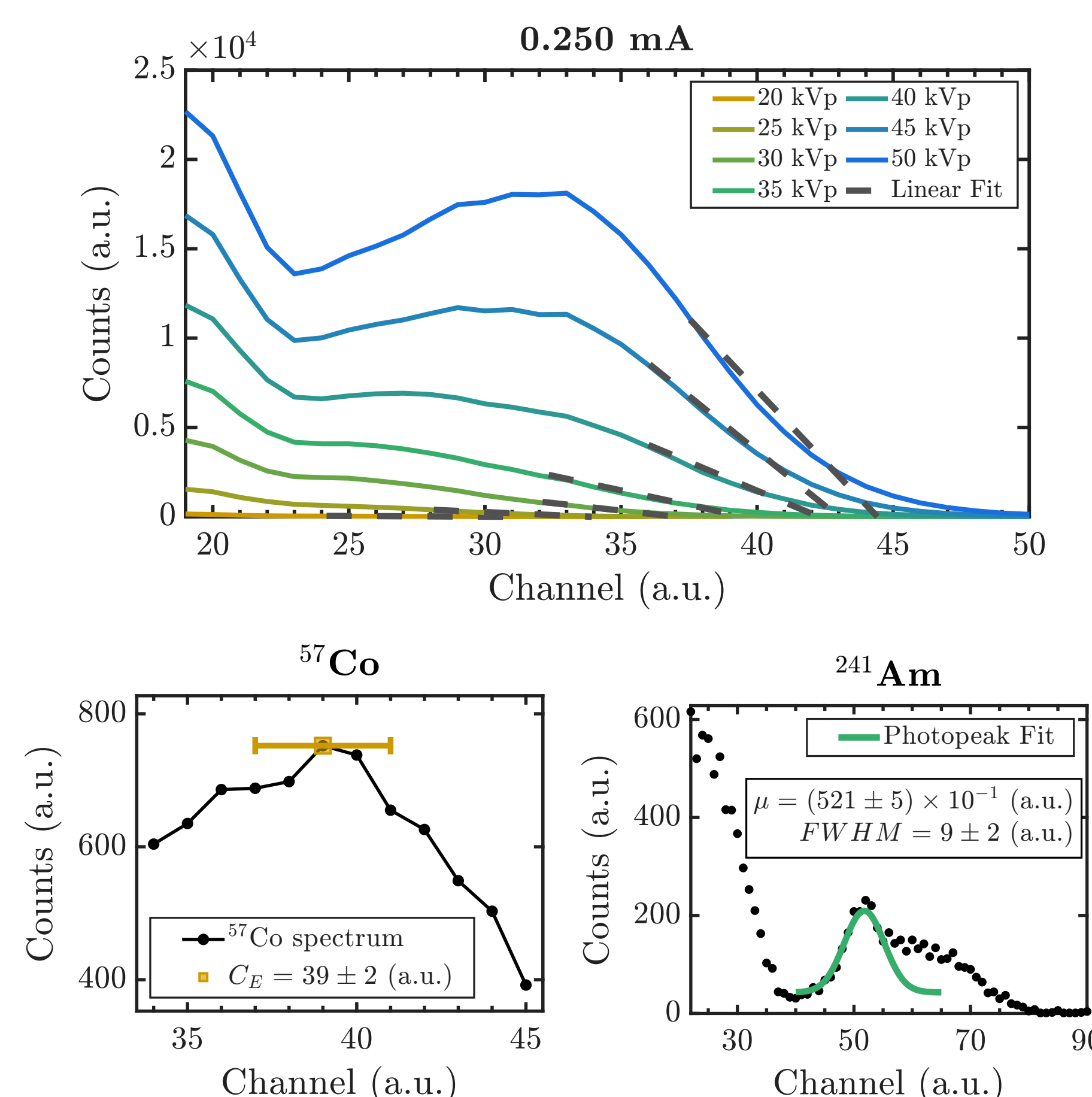


PROTOTYPE

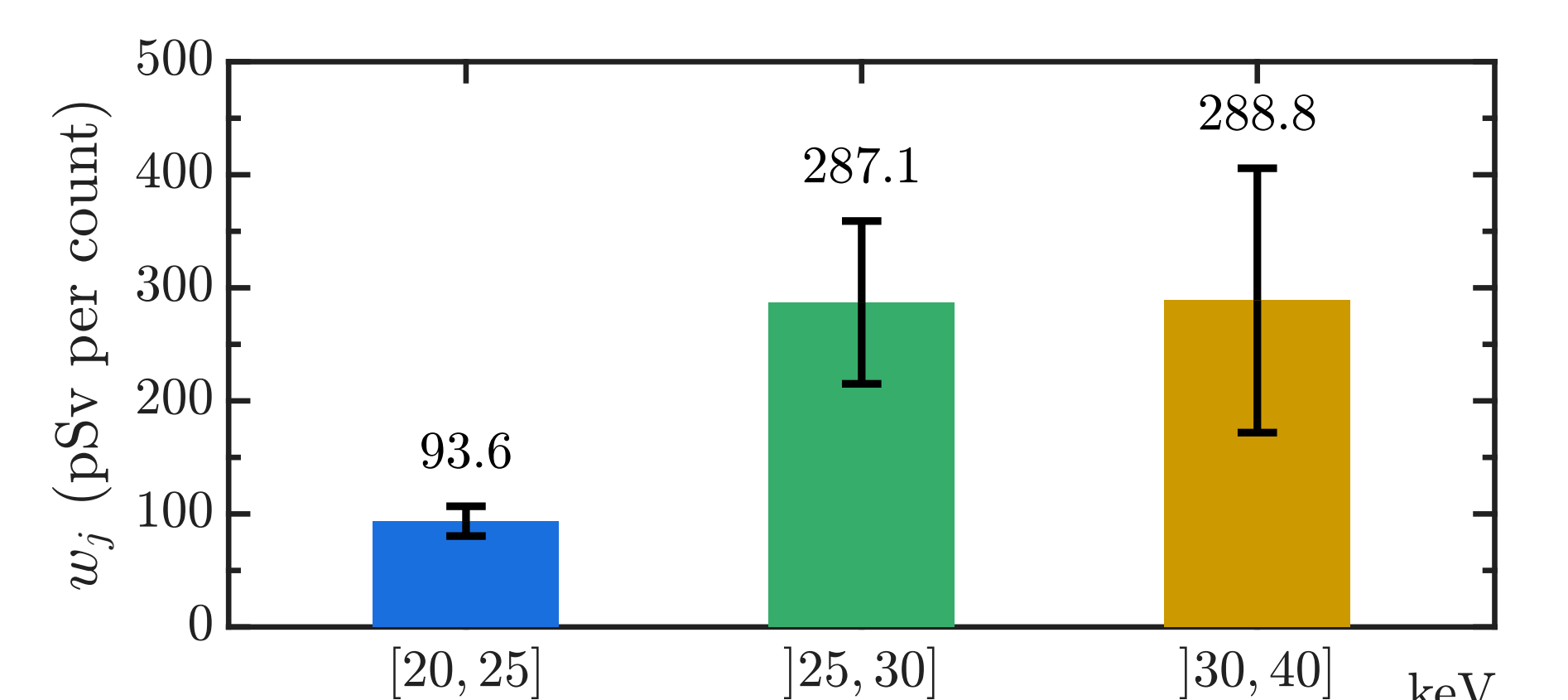
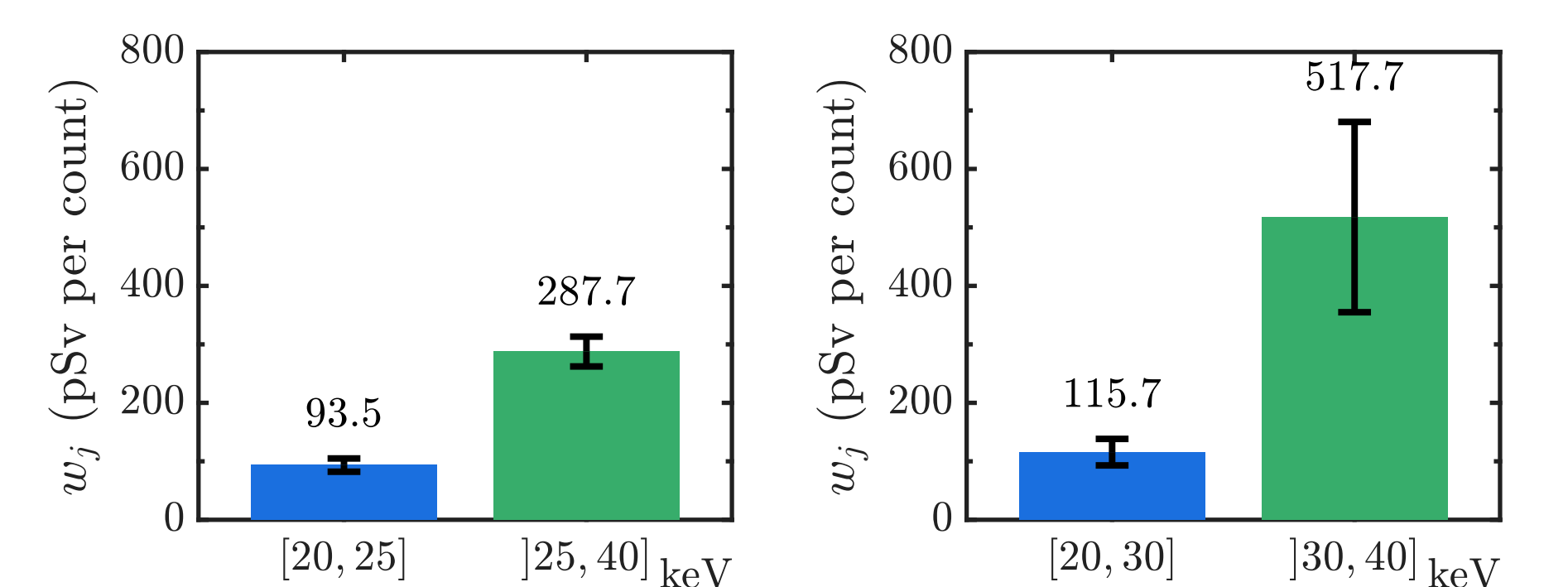
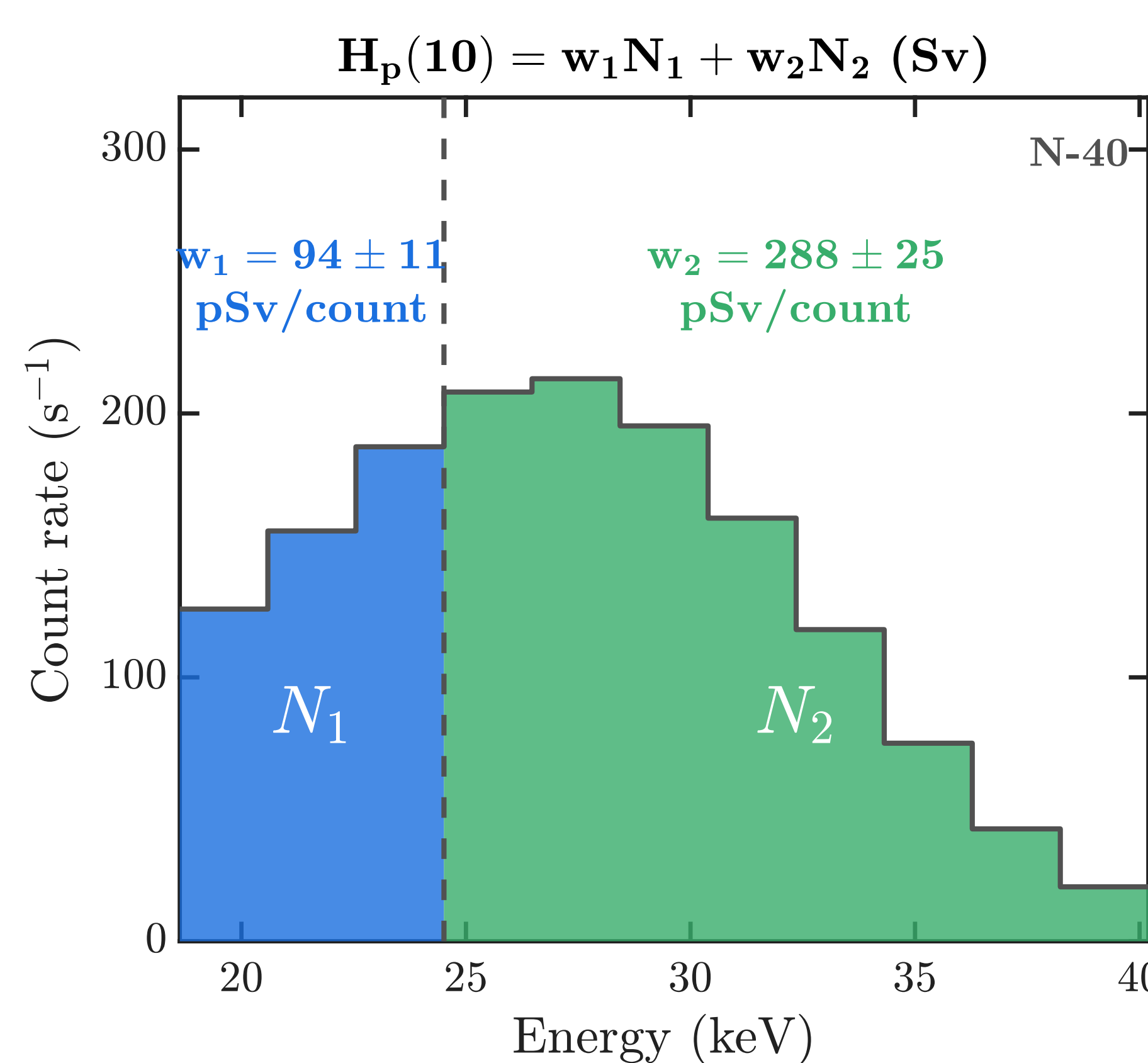


Parameter	Value
Dead time	$17.8 \pm 0.4 \mu\text{s}$
Calibrated energy range	20 to 60 keV
Operational unit	$H_p(10)$
Calibrated dose range	N-25 to N-40 0.063 to $101 \mu\text{Sv/h}$
Dimensions	$6.3 \times 7.2 \times 2.5 \text{ cm}$
Mass	141 g
Autonomy	8.6 h

RESULTS



Linear channel response between 20 and 60 keV



Two-bin model gives the best fit to all three qualities

CONCLUSIONS

A functional, plastic scintillator-based, wireless active personal dosimeter was built and characterised. The scintillator can be calibrated in both energy and dose, despite its known poor energy resolution. Future work will extend both calibrations to a wider energy range and to more reliable dose points, reduce the size of the unit, and scale it to monitor several anatomical regions, such as the chest, eyes, thyroid and hands, in real-time.

REFERENCES

- <https://doi.org/10.1259/bjr.20210436>
- <https://doi.org/10.1007/s00330-025-11566-5>
- ISO 4037-1:2019, *Radiation characteristics and production methods*.
- ISO 4037-3:2019, *Calibration of area and personal dosimeters*.

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